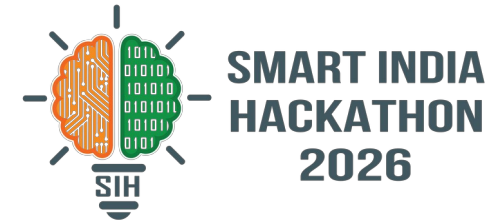


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID - 26172
- Problem Statement Title - Low Latency and Efficient Voice Activator for Edge Devices
- Theme - Miscellaneous
- PS Category - Hardware
- Team ID -
- Team Name (Registered on portal) - DRAMAS

PRAHARI

प्रहरी • "sentinel"

Privacy-preserving Real-time Acoustic Hybrid Activation for Resource-constrained IoT

Hears the Hindi keyword on a low-cost ESP32-S3, decides on the chip, and streams to cloud speech recognition over a link that is **already open**. No audio leaves the device before the keyword.

96.4 %

real-speaker
detection

0.48 / h

false wakes
(bar < 0.5 / h)

50 ms

keyword → AWS
round trip, 4G

7.8 %

idle CPU
(limit 10 %)

195 KB

peak RAM
(limit 256 KB)



PROPOSED SOLUTION

A complete voice-activation device: a **20.7 KB neural network on a low-cost ESP32-S3** listens for the Hindi keyword **प्रहरी**, decides **on the chip**, and hands the speech to a cloud recogniser over a link that is **already open**. **No audio leaves the device before the keyword.**

★ KEY FEATURES — WHAT IT DOES, AND THE MEASUREMENT THAT PROVES IT ★ unique to PRAHARI · tag = PS metric it serves



On-chip wake word

A tiny int8 neural network decides on the microcontroller — no network needed.

EFFICIENCY
20.7 KB model · 27.2 ms


Privacy by design

No audio leaves before प्रहरी; a privacy LED lights while audio streams.

PRIVACY
0 bytes before keyword


★ Link already open

Connected at boot and kept alive — no handshake after the keyword.

LATENCY
1 ms · 50 ms RTT to AWS


★ Zero-clip pre-roll

The last 1.5 s of audio waits compressed in RAM, so no syllable is ever cut.

LATENCY
1.5 s held in 12.2 KB


Sleeps while it listens

Energy gate + dual-rate scan: the network runs once a second in silence.

EFFICIENCY
7.8 % idle CPU (limit 10 %)


★ Burst scanning

When a sound looks like the keyword, it zooms in with 40 ms scans.

ACCURACY
+9.7 pts for +0.4 % CPU


Rejects look-alike words

Tells प्रहरी from प्रहर, पहरी, प्रहार — one ~100 ms vowel apart.

ACCURACY
0 / 40 real sound-alikes


Near-zero false wakes

Trained against 605 h of real Indian speech, TV, music and noise.

ACCURACY
0.48 / h · 0/16 real use


Knows when you stop

On-device voice-activity detection closes the stream when speech ends.

EFFICIENCY
ends ~0.8 s after speech


4× less bandwidth

IMA-ADPCM compression keeps speech clear for the recogniser.

EFFICIENCY
64 kbps · 24.8 dB SNR


Self-healing link

Keep-alive, auto-reconnect, LAN auto-discovery; the mic never pauses.

RELIABILITY
rode out a 1.2 s 4G stall


Self-test & field tuning

Bit-exact model check at every boot; threshold and gain tunable live.

RELIABILITY
no reflash to re-tune

HOW IT MEETS PS 26172

✓ Custom keyword
प्रहरी, from scratch

✓ Open-source only
no voice SDK

✓ RAM < 256 KB
195 KB peak

✓ Idle CPU < 10 %
7.8 % measured

✓ Near-zero false wakes
0.48 / h

✓ Low latency
50 ms RTT to AWS

✓ Real MCU
XIAO ESP32-S3

TECHNOLOGIES USED

HARDWARE

XIAO ESP32-S3 PDM mic 256 KB budget

EDGE FIRMWARE

C/C++ ESP-IDF TFLite Micro ESP-NN

ML & TRAINING

Python TensorFlow TFLite int8 Keras

LINK & CLOUD

ADPCM warm TCP Vosk ASR AWS EC2

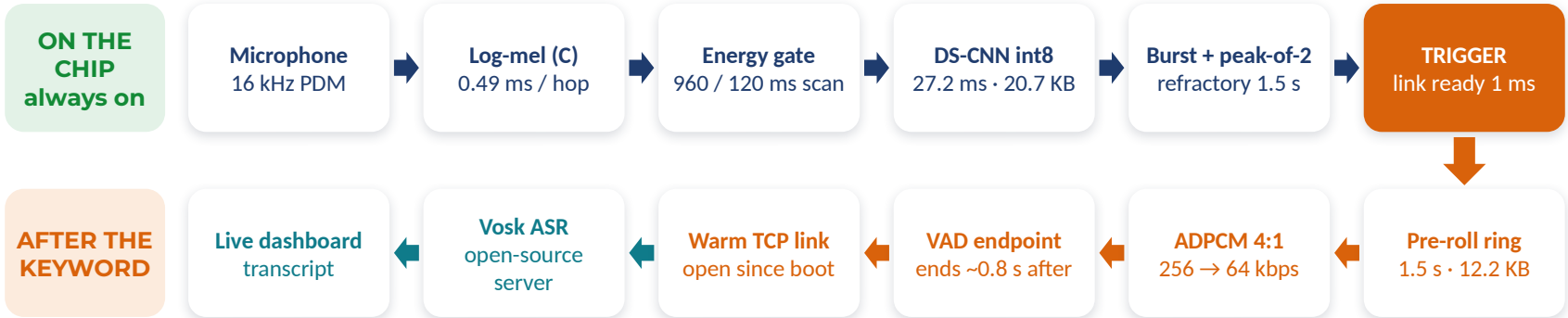
TOOLS WE BUILT

bench_log fa_eval retrain arena_calc

ENGINEERING HIGHLIGHTS

- Bit-exact laptop ↔ chip: 0 of 94,080 features differ
- Real-input FFT: 512-pt done as 256-pt complex
- ESP-NN kernels 15.6× faster than reference
- Exact graph surgery: MEAN → AvgPool, 32× faster
- Digital twin of the board's scan scores every epoch

METHODOLOGY AND PROCESS FOR IMPLEMENTATION — WORKING PROTOTYPE

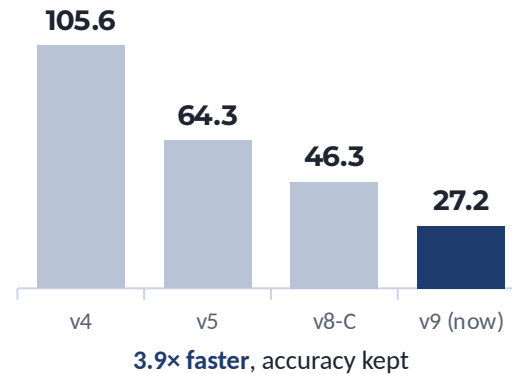


To AWS Mumbai over 4G: link ready 1 ms · device side ≤ 4 ms · round trip 50 ms median · survives a 1.2 s network stall

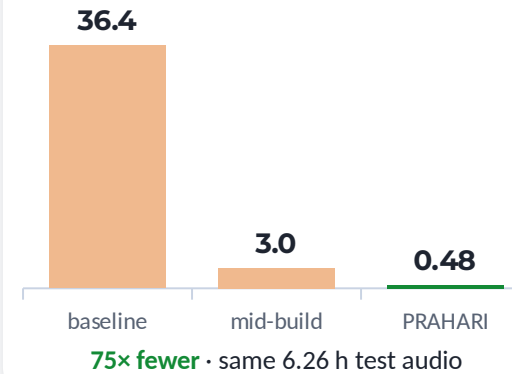
MODEL

Input 98×40 log-mel (≈ 1 s) → Conv 2×16 → 4 depthwise-separable blocks (16 / 32 / 32 / 32 ch) → AvgPool → sigmoid
20.7 KB int8 · 17 operators · 36 KB tensor arena · BatchNorm folded · bit-exact with training

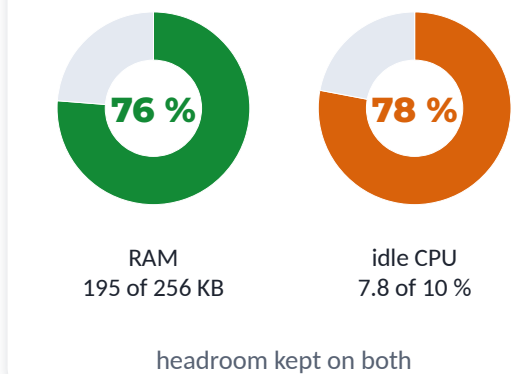
Inference on the chip (ms)



False wakes per hour



Problem-statement budget used



✓ ANALYSIS OF THE FEASIBILITY — ALREADY BUILT, RUNNING AND MEASURED

TESTED ON REAL HUMAN VOICES — BY GENDER				
fresh recordings, scored the way the board scans				
SPEAKERS	TAKES	DETECTED @ 0.85 (shipped)	DETECTED @ 0.70	SOUND-ALIKES
Female	69	94.2 %	100 %	0 / 20
Male	70	98.6 %	98.6 %	0 / 20
Both	139	96.4 %	99.3 %	0 / 40

10 / 10

captures clean,
0 ms audio dropped

100 %

of 340 held-out
synthetic test keywords

Bit-exact

chip reproduces the
laptop model exactly

CHALLENGES AND RISKS → HOW WE OVERCAME THEM · one row per aspect, everything about it in one place

ASPECT	BUILT & MEASURED	CHALLENGE WE HIT	HOW WE OVERCAME IT	NEXT (FINALS)
LATENCY	Warm link from boot, 1.5 s pre-roll 1 ms ready · 50 ms RTT to AWS	Cold TLS handshake: 1,210 ms	All connection set-up moved to boot	TLS on the warm link
IDLE CPU	Energy gate + dual-rate scan 7.8 % idle (limit 10 %)	Front-end alone cost 17.6 %	Real FFT, -O3, ring buffer: 0.49 ms / hop	Power profile (INA226)
MEMORY	int8 model, 36 KB arena, PSRAM off 195 KB of 256 KB · model 20.7 KB	First model needed a 76 KB arena	ESP-NN-aligned channels → 36 KB arena	Keep ≥ 60 KB headroom
SPEED	17-op DS-CNN on ESP-NN kernels 27.2 ms per decision	First model took 105.6 ms	6 architectures profiled per operator → 3.9×	Streaming inference
FALSE WAKES	605 h real Indian audio as negatives 0.48 / h · 0 / 16 in real use	First model: 36.4 / h	Breadth, not phonetics → 75× fewer	Long field logs
DETECTION	Real speakers 96.4 % (F 94.2 · M 98.6) 100 % of 340 synthetic test keywords	Almost no recorded प्रहरी speech exists	Own multi-generator voice corpus, audited by ear	Consented speaker corpus

POTENTIAL IMPACT ON THE TARGET AUDIENCE



ISRO ground stations & cleanrooms

Hands-free commands with gloves or PPE on; audio can stay on-premise.



Field & rural sites

Detection needs no network; the uplink runs on a 4G hotspot at 64 kbps.



Wearables & safety devices

A distress keyword that listens all day at 7.8 % CPU on a small MCU.



Indian IoT & smart-home makers

A licence-free Hindi wake word in 20.7 KB — the chip stays free.



Accessibility

Hands-free control in Hindi for the elderly and people with limited mobility.



Defence & government

An auditable, fully open Indian stack — no foreign SDK.

BENEFITS OF THE SOLUTION — SOCIAL, ECONOMIC, ENVIRONMENTAL, STRATEGIC



SOCIAL & PRIVACY

- No audio leaves before the keyword
- Privacy LED lit whenever audio streams
- Hindi keyword, voices from 9 Indian languages



ECONOMIC

- No per-user wake-word SDK fees
- 4× less audio data on cellular links
- Commodity ESP32-class chips; ASR on a laptop-class CPU



ENVIRONMENTAL

- 3.5× less idle compute than always-on scanning
- No datacentre streaming for a rare keyword
- 4× less data → less radio-on time



STRATEGIC

- Fully open, auditable Indian voice stack
- Built on Indian open speech data (AI4Bharat)
- Portable C core — no vendor lock-in



IMPACT IN NUMBERS — BEFORE → AFTER (MEASURED)

False wakes per hour (same 6.26 h of audio)

36.4

➔ **0.48**

75× fewer

CPU while listening (scan always → PRAHARI idle)

27.7 %

➔ **7.8 %**

3.5× less

Keyword → link ready (cold TLS → warm link)

1,210 ms

➔ **1 ms**

no handshake

Audio per stream (raw PCM → ADPCM)

256 kbps

➔ **64 kbps**

4× less data



DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

OUR RESEARCH — WHAT WE PROVED ON SILICON

1

Breadth beats phonetics

Adding 605 h of real Indian speech cut false wakes 75x; phonetic hard-negative mining gave no measurable gain.

2

Reverb can erase the keyword

प्रहरी differs from प्रहर by a ~100 ms final vowel; reverb above RT60 0.35 s smears it, so we cap it.

3

Kernel dispatch decides memory

ESP-NN takes a slow path unless channels are multiples of 8; matching them cut the arena 76 → 36 KB.

4

The scan schedule is part of accuracy

A slow scan can step over a short score peak; burst scanning catches it for +0.4 % compute.

5

Bit-exact from Python to silicon

Our C front-end matches training on all 94,080 int8 values — the laptop result is the chip result.

KEYWORD-SPOTTING RESEARCH

- Zhang et al., “Hello Edge: KWS on Microcontrollers”, 2017 — arxiv.org/abs/1711.07128
- Choi et al., “Temporal Convolution for Real-Time KWS”, 2019 — arxiv.org/abs/1904.03814
- Rybakov et al., “Streaming KWS on Mobile Devices”, 2020 — arxiv.org/abs/2005.06720
- Kim et al., “Broadcasted Residual Learning for KWS”, 2021 — arxiv.org/abs/2106.04140
- Banbury et al., “MLPerf Tiny Benchmark”, 2021 — arxiv.org/abs/2106.07597
- Park et al., “SpecAugment”, 2019 — arxiv.org/abs/1904.08779

ON-DEVICE STACK & WAKE-WORD PRACTICE

- David et al., “TensorFlow Lite Micro”, MLSys 2021 — arxiv.org/abs/2010.08678
- Espressif ESP-NN optimised int8 kernels — github.com/espressif/esp-nn
- openWakeWord — “< 0.5 false accepts / hour” bar — github.com/dscripka/openWakeWord
- microWakeWord on ESP32-S3 — home-assistant.io/blog/2024/02/21/voice-chapter-6
- Vosk offline speech recognition — alphacephei.com/vosk

DATASETS & VOICES WE TRAINED ON

- IndicVoices — arxiv.org/abs/2403.01926 · Kathbath — arxiv.org/abs/2208.11761
- Shrutilipi — arxiv.org/abs/2208.12666 · OpenSLR 104 — openslr.org/104
- Common Voice — arxiv.org/abs/1912.06670 · MUSAN — arxiv.org/abs/1510.08484
- FSD50K — arxiv.org/abs/2010.00475 · ESC-50 — github.com/karolpiczak/ESC-50
- IndicF5 — github.com/AI4Bharat/IndicF5 · edge-tts — github.com/rany2/edge-tts

OPEN SOURCE & REPRODUCIBLE — firmware, training pipeline, server and board logs: github.com/Rkaid09/PRAHARI · every device number here regenerates from logs with bench_log.py and fa_eval.py